Faisal Ahmed Chowdhury

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Faisal Ahmed Chowdhury | Faisal-A-Chowdhury | Faisal Ahmed Chowdhury

Beaumont, Texas, USA

Objective

I am a first-year M.Sc. student in Electrical and Computer Engineering with a strong interest in quantum computing and quantum information science. My current research focuses on exploring different quantum error correction (QEC) techniques to improve the reliability of noisy quantum systems. My interests span both device-level and algorithmic development, including quantum communication, quantum machine learning, quantum-safe security, and hybrid quantum–classical computing, particularly exploring how quantum principles can be applied to improve non-volatile memory systems and RAM architectures for next-generation computing. I am especially interested in bridging machine learning, quantum information, and modern computing architectures to address challenges in reliable quantum computation and secure communication. With a long-term goal of pursuing an academic career, I aim to contribute impactful research and develop practical quantum technologies.

Research Interest

- **Quantum Error Correction**
- **Quantum Algorithms**
- **Quantum Communication**
- **Quantum Security**
- **Quantum Machine Learning**
- **Computer Architecture**
- **Quantum Photonics**
- **Memory Systems (NVM, RAM)**
- **Electronics & Embedded Systems**
- **Optoelectronic Devices**

Education

- **Lamar University** — *Jan 2026 - Ongoing*
  M.Sc. in Electrical Engineering — Texas, USA
  Department of Electrical and Computer Engineering
- **Chittagong University of Engineering and Technology** — *Feb 2017 - Aug 2022*
  B.Sc. in Electrical and Electronic Engineering — Chittagong, Bangladesh
- **Chittagong College** — *July 2014 - May 2016*
  Higher Secondary School Certificate — Chittagong, Bangladesh

Experience

- **Graduate Research Assistant** — *Jan 2026 - Ongoing*
  Lamar University, Department of Electrical and Computer Engineering — Texas, USA
  - Working on quantum error correction and hybrid classical–quantum system development.
  - Assisting in experiment design, simulation, and manuscript preparation for high-impact journals.

Relevant Courses

- Fundamentals of Quantum Computing
- Analysis of Algorithms
- Artificial Intelligence
- Advanced VLSI Design & Testing
- Cyber-Physical Systems & Security
- VLSI Circuits & Design
- Microprocessors & Embedded Systems
- Digital Signal Processing
- Digital Electronics
- Control Systems
- Numerical Techniques
- Telecommunication Engineering

Skills

- **Programming Languages:** Python, MATLAB, Verilog, Assembly Language.
- **Simulation Software:** COMSOL Multiphysics, Cadence, Simulink, PSpice, Proteus.
- **Data Science & Machine Learning:** TensorFlow, PyTorch, Scikit-learn, NumPy, Pandas, Matplotlib.
- **Other Tools & Technologies:** OpenCV, Qiskit, Pennylane, LaTeX.
- **Soft Skills:** Organizing, Collaborating, Teaching, Academic Writing.

Academic Projects

P=Project

[P.1] **Physical design** of a microcontroller using CADENCE INNOVUS Implementation System.

[P.2] **Configurable Logic Block (CLB)** design using CADENCE Virtuoso.

[P.3] **Parkinson's Disease Detection** using Voice Analysis.

[P.4] **Privacy Preserving GPS** implementation using STM32.

[P.5] **Robotic ARM** Design and Simulation.

PUBLICATIONS

C=CONFERENCE, J=JOURNAL

[C.1] Disagreement-Gated Graph-Residual Correction for Mismatched MWPM Decoding in Surface Codes
(Presented in IEEE Quantum Week 2026, Workshop Track) [PDF]

- Proposed a disagreement-gated graph-residual decoder that selectively corrects mismatched MWPM errors.
- Evaluated on surface codes with $d = 3, 5, 7$, achieving improved LER and positive net correction.

[C.2] Benchmarking Quantum Image Dataset Transform Pre-processing on Quantum Architectures
(Presented in IEEE Quantum Week 2026, Workshop Track) [PDF]

- Benchmarked QIDT-based image pre-processing across QCNN, QuanvNet, and QViT architectures.
- Evaluated accuracy, F1-score, throughput, and energy across MNIST, Fashion-MNIST, and MedMNIST.

[C.3] Machine Learning–Enhanced Adaptive Fidelity Optimization in W-State Quantum Teleportation
(Presented in QCUF 2026 at Oak Ridge National Laboratory and got Best Poster Award) [Preprint] [Ref]

- Developed an ML-driven adaptive correction framework for W-state quantum teleportation.
- Improved fidelity by 44.81% and validated performance on IBM Quantum hardware.

[C.4] Recurrent Neural Network Architecture to Predict Supercontinuum Generation in Chalcogenide- Silica Hybrid Photonic Crystal Fiber
(Accepted and published at the IEEE COMPAS 2024 Conference) [DOI] [PDF]

- Proposed an RNN model for rapid supercontinuum prediction in chalcogenide–silica PCF, outperforming COMSOL speed.
- Achieved high accuracy with a mean square error of 0.00078784.

[C.5] Highly Sensitive Dual-Channel PCF Based SPR Biosensor for Multi-Analyte Detection with Fabrication Tolerance Analysis
(Accepted and published at the IEEE ICECE 2024 Conference) [DOI] [PDF]

- Designed a dual-channel PCF-SPR sensor using Au and Ta_2O_5.
- Achieved up to 83,000 nm/RIU sensitivity and 1.2×10^{-6} RIU resolution.

[C.6] A Novel C-Grooved SPR PCF Biosensor for Early Detection of Malaria Using Fuzzy Logic
(Accepted and published at the IEEE ICRPSET 2024) [DOI] [PDF]

- Developed a C-grooved SPR PCF sensor with fuzzy logic for malaria phase detection, achieving high sensitivities (1704 RIU^{-1}, 15,300 nm/RIU).
- Optimized plasmonic performance with dual C-grooves and Au/TiO_2 layers, enabling selective detection of healthy and malaria-infected RBCs.

[J.1] Machine Learning-Assisted Highly Sensitive K-Shaped Core PCF-SPR Sensor for Intelligent Refractive Index Sensing
(Under peer review in Advanced Sensor Research)[Preprint]

JOB EXPERIENCE

- **BSRM Steels Limited [🌐]** *Feb 2023 - March 2024*
  Electrical Maintenance and Automation Engineer Chittagong, Bangladesh
  - Testing of electrical equipment, fault analysis, and maintenance.
  - Installation and calibration of electrical equipment.
  - PLC automation system control and fault analysis.

REFERENCES

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Lamar University
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Dr. Mohammad Istiaque Reja
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